

Effect of non-oil-seed pulses on glycaemic control: a systematic review and meta-analysis of randomised controlled experimental trials in people wi...

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AIMS/HYPOTHESIS: Dietary non-oil-seed pulses (chickpeas, beans, peas, lentils, etc.) are a good source of slowly digestible carbohydrate, fibre and vegetable protein and a valuable means of lowering the glycaemic-index (GI) of the diet. To assess the evidence that dietary pulses may benefit glycaemic control, we conducted a systematic review and meta-analysis of randomised controlled experimental trials investigating the effect of pulses, alone or as part of low-GI or high-fibre diets, on markers of glycaemic control in people with and without diabetes. METHODS: We searched MEDLINE, EMBASE, CINAHL, and the Cochrane Library for relevant controlled trials of ≥ 7 days. Two independent reviewers (A. Esfahani and J. M. W. Wong) extracted information on study design, participants, treatments and outcomes. Data were pooled using the generic inverse variance method and expressed as standardised mean differences (SMD) with 95% CIs. Heterogeneity was assessed by chi (2) and quantified by I (2). Meta-regression models identified independent predictors of effects. RESULTS: A total of 41 trials (39 reports) were included. Pulses alone (11 trials) lowered fasting blood glucose (FBG) (-0.82, 95% CI -1.36 to -0.27) and insulin (-0.49, 95% CI -0.93 to -0.04). Pulses in low-GI diets (19 trials) lowered glycosylated blood proteins (GP), measured as HbA(1c) or fructosamine (-0.28, 95% CI -0.42 to -0.14). Finally, pulses in high-fibre diets (11 trials) lowered FBG (-0.32, 95% CI -0.49 to -0.15) and GP (-0.27, 95% CI -0.45 to -0.09). Inter-study heterogeneity was high and unexplained for most outcomes, with benefits modified or predicted by diabetes status, pulse type, dose, physical form, duration of follow-up, study quality, macronutrient profile of background diets, feeding control and design. CONCLUSIONS/INTERPRETATION: Pooled analyses demonstrated that pulses, alone or in low-GI or high-fibre diets, improve markers of longer term glycaemic control in humans, with the extent of the improvements subject to significant inter-study heterogeneity. There is a need for further large, well-designed trials.